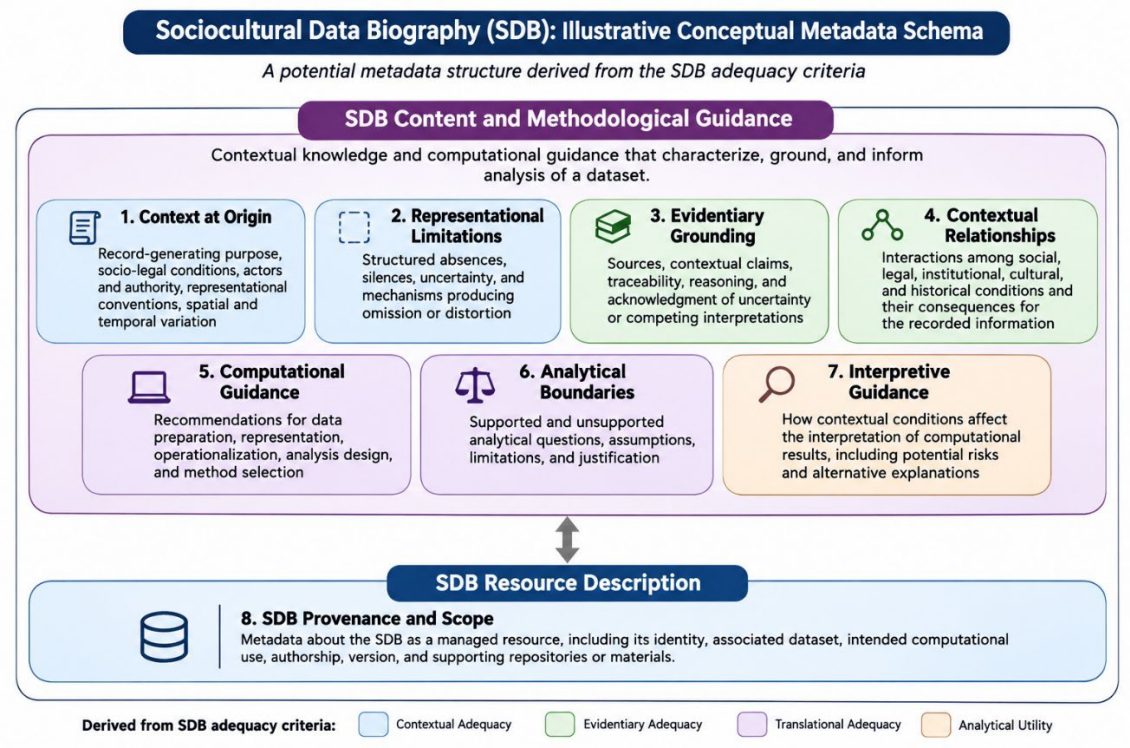
Sociocultural Data Biography (SDB):
Preliminary Conceptual Metadata Schema

This preliminary conceptual metadata schema illustrates one possible representation of the Sociocultural Data Biography (SDB), the knowledge resource produced through Sociocultural Contextualization for Computational Treatment (SCCT). The schema is derived from the four SDB adequacy criteria—Contextual Adequacy, Evidentiary Adequacy, Translational Adequacy, and Analytical Utility—and their operationalization in the archival use case. It proposes seven elements for representing sociocultural contextual knowledge, its evidentiary grounding, and its implications for computational treatment and interpretation, together with an eighth element describing the SDB as a managed information resource. The proposed elements and attributes are intended to support explicit relationships among sociocultural conditions, recorded information, supporting evidence, and computational guidance.



High-level element	Illustrative metadata attributes	Principal adequacy relationship
1. Context at Origin	Record-generating purpose; institutional mandate; socio-legal structures; actors and authority; representational conventions; spatial and temporal variation	Contextual
2. Representational Limitations	Absence type; affected attributes; mechanisms producing absence; uncertainty; representational bias or distortion	Contextual, Evidentiary
3. Evidentiary Grounding	Contextual claim; source reference; evidence type; supporting observations; interpretive reasoning; uncertainty or competing interpretations	Evidentiary
4. Contextual Relationships	Related contextual conditions; type of relationship; combined effects on recorded information; supporting evidence	Contextual, Evidentiary
5. Computational Guidance	Affected data attributes; decision category; recommended treatment; justification; constraints or alternatives	Translational
6. Analytical Boundaries	Supported analytical questions; unsupported or unreliable questions; assumptions; limiting conditions; justification	Translational, Analytical Utility
7. Interpretive Guidance	Relevant contextual conditions; anticipated interpretive risks; guidance for evaluating computational findings; anomalous-pattern explanations	Analytical Utility
8. SDB Provenance and Scope	SDB identifier; associated dataset; intended computational use; creator; date/version; supporting repositories	Resource management (proposed)

This schema is an illustrative conceptual design rather than a formal metadata specification or validated implementation. Consistent with the representation-independent definition of the SDB, it does not prescribe a particular encoding format, vocabulary, or technical architecture. Further development would require refinement of element definitions and relationships, specification of metadata properties and constraints, alignment with relevant metadata standards and ontologies, and evaluation of the schema's completeness, usability, and applicability across domains. In particular, future work should investigate whether structured representations of contextual claims, supporting evidence, and computational guidance can preserve the traceability and methodological functions required of an adequate SDB.